Misty Rainforest – 2D Jump & Run

https://github.com/rstoiberer/CMS320-Fall2025

CMS 320 – Video Game Design

Instructor:

Dr. Sirazum Munira Tisha

Team Members & Roles:

Sydney Eckstein: Artist / Audio & UI Designer

Stella Fruijtier: Gameplay programmer / Technical Lead

Richard Stoiberer: Level & Game Design / General Project Manager

December 11, 2025

Introduction

a. Background & Motivation

Misty Rainforest is a 2D jump-and-run adventure following Komea, a human turned skeleton who must journey through a fog-covered rainforest to recover a sacred pearl guarded by merfolk, a tribe of mythical, aquatic humanoids. If Komea secures the pearl, his curse breaks, and he regains his human form. We chose this concept because we wanted to create a concise fantasy experience that focuses on pacing, ambiance, and replayability.

b. Target users

The game targets casual platformer fans, ages 8 and up, who enjoy short indie experiences and fantasy action. We designed the game to run both Windows PC and Apple Mac, making it accessible to most casual players.

c. Game World / Context

The setting is a spooky-calm rainforest with low fog and constant rain ambience. Mythical merfolk protect the treasure that Komea seeks. The journey gradually transitions from gentle exploration to enemy-full caverns where the player is ultimately met with a gatekeeper in the third and final level before arriving at the sacred altar where the pearl resides.

d. Game Rules

- Defeat all merfolk on screen before advancing towards the right side where the gate opens
- One hit to Komea (player) results in death, restars at level 1
- The only hazard besides enemies and falling off the plarforms is water that slows movement
- Score is based on completion time, encouraging replayability

e. Controls

Move (run): left arrow/right arrow

Jump: space bar

Spear attack: J

Interact: E

These controls are optimized for keyboard play, keeping gameplay simple and intuitive.

f. Sketch / Visual plan

The sketches of the levels of Misty Rainforest can be found on GitHub, in the document titled "MistyRainforestLevelSketches"

g. Task distribution

Sydney Eckstein: sprites, background, music, sound effects, animations

Stella Fruijtier: core mechanics, physics game system development

Richard Stoiberer: level & game design, project vision, core mechanics

Game Development

Core game mechanics & systems programming: Collaboration Stella Fruijtier & Richard Stoiberer.

- Player movement & physics
    - Implemented responsive running and jumping behavior using Unity's Rigidbody 2D
    - Tuned acceleration, gravity scale, jump impulse, and air control for balanced platforming
- Combat & Spear attack
    - Built spear hit detection using raycasts and colliders
    - Developed merfolk attack reach, timing, and cooldown
- Enemy behavior:
    - Scouting: simple movement left to right

    - Attack: Komea is detected within a certain distance, kills if directly touched
- Water hazard mechanics:
    - Player movement speed and jump ability are reduced when entering water zones
- Death & Restart system
    - One-hit death system resets player to Level 1
    - Komea dies if falls below lowest tiles or gets hit by Merfolk
- Timer & Score tracking
    - Score = total completion time
    - Display final time at game completion for replayability
    - Score timer keeps running until completed the game

Division of work within this collaboration:

System	Primary Developer	Support/Testing
Player movement mechanics	Shared	Shared
Enemy logic	Richard	Stella
Comabat mechanics	Richard	Stella
Water hazard system	Stella	Richard
Death/Restart	Stella	Richard
Timer	Shared	Shared

<u>Level Design & Game Pacing</u>: Lead Richard, Playtesting shared with Stella:

Richard led the design of each level's layout, difficulty curve, and pacing. He created the environmental flow with difficulty progression.

- Level 1 – Xilan Jungle

- Few merfolk
- Wide platform, slow pacing
- Designed to teach jmp & attack mechanics without pressure

- Level 2 – River Caverns
  - Introduced water hazards strategically to slow movement
  - Enemy density increases to stress timing decisions
- Level 3 – Orasc Ascend
  - Narrow timing windows to attack
  - Final boss challenge
  - No lower wider platform to catch on if fall

Stella supported by:

- Playtesting movement on platforms to decide enemy density
- Adjusting physics values and enemy hitboxes to prevent unfair challenge

<u>Art design, Animation & Audio</u>: Lead Sydney Eckstein

Sydney defined the game's visual identity and audio environment.

- Character & enemy:
  - Designed and hand-drew all sprites with animated states (idle, walk, jump, attack, death)
  - All drawings were made on Goodnotes using an iPad and Apple Pencil, Krita was later incorporated for more advanced designs
  - Implemented all animations using animator controllers and animation clips
- Environmental tiles & background art:
  - Manually drew all backgrounds and tile sets

- Bound each drawing to canvas objects

- Sound & Music:
  - Implemented all background music, attack & death sound effects through scene loads and buttons
  - Recorded original death sound effect using Voice Memos
  - Balanced sounds volume, delay and duration
  - Created an ambience
- UI:
  - Designed menu, backstory, endscene, and credits scene
  - Synced animation states with gameplay triggers provided by Stella & Richard

Collaboration workflow summary:

Task	Main Contributors
Game mechanics	Stella & Richard
Level layout & difficulty	Richard (+ Stella)
Art, Audio, Animations	Sydney
Integration & Testing	All members

Final Game

Link to the gameplay: https://github.com/rstoiberer/CMS320-Fall2025/releases/tag/Gameplay

How to watch gameplay recording: GitHub repository, releases, download MistyRainforestGameplay.mov

Discussion and future work

Developing Misty Rainforest was both a creative and technical experience. From the beginning, our goal was to build a 2D platformer that combined simple controls with a visually rich environment. As a team, we had a clear direction, but turning that vision into a functional Unity game required patience and dedication. Learning how animations, combat mechanics, prefabs, art, and UI all interact inside the project made us understand how each detail contributes to the overall player experience.

One of the biggest challenges we faced was keeping the game consistent while programming new features. Often, when we implemented something new, something else would unexpectedly break. For example, adding the platform art suddenly made the character not able to jump anymore. Later, adjusting the water platforms made the enemies fall off when starting the game. To solve these issues, we learned to test more frequently, organize our prefabs better, and avoid changing objects without making sure that the rest was working. This taught us how important version control and prefab testing are in Unity.

From a creative standpoint, it felt very empowering to take ownership of the look and feel of a game from sketch to final product. One of the biggest challenges faced was solidifying the desired art style and methodology for ensuring reproducibility with each animation state for a smooth and seamless gameplay experience. The way in which we overcame this challenge was by doing extensive research on the drawing capabilities of Goodnotes and Krita and devised a strategy based on how they both best complement each other. Art was drawn by hand using an iPad and Apple pencil on Goodnotes, then uploaded to a MacBook for final details using Krita. Finally, the white background was removed using a free online background remover which was then uploaded to Unity.

In the future, we would like to add more functionality to Komea and the merfolk. We would include increasingly complex attacking features and defense mechanisms (utilizing

Komea's shield, ducking, etc) that allows for more sophisticated combat and added challenge for users. In terms of audio, implementing distinctive sound effects that would better identify Komea and the merfolk respectively would complete well-rounded character profiles and enhance the user experience as they would receive more descriptive feedback on their gameplay.